

Complete Course: Probability and Statistics

From Basic Concepts to Advanced Inference

Mathematical Foundations for Data Science

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1 Introduction to Probability Theory

1.1 Historical Context

Probability theory originated from the study of gambling games in the 17th century, with contributions from Pascal, Fermat, Bernoulli, and Laplace. Modern probability theory was formalized by Kolmogorov in the 1930s.

1.2 Basic Probability Concepts

Definition 1.1 (Sample Space). The **sample space** Ω is the set of all possible outcomes of a random experiment.

Definition 1.2 (Event). An **event** A is a subset of Ω ($A \subseteq \Omega$).

1.3 Kolmogorov Axioms

Definition 1.3 (Probability Measure). A function $P : \mathcal{F} \rightarrow [0, 1]$ on a σ -algebra $\mathcal{F}$ of subsets of Ω is a probability measure if:

1. $P(\Omega) = 1$
2. $P(A) \geq 0$ for all $A \in \mathcal{F}$
3. For countable disjoint events $A_1, A_2, \dots$:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

1.4 Basic Probability Rules

Theorem 1.4 (Probability Rules). 1. $P(\emptyset) = 0$

2. $P(A^c) = 1 - P(A)$
3. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
4. If $A \subseteq B$, then $P(A) \leq P(B)$
5. $P(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n P(A_i)$ (Boole's inequality)

Proof of Rule 3. Let $B_1 = A \cap B^c$, $B_2 = A \cap B$, and $B_3 = A^c \cap B$. These are disjoint and:

$$\begin{aligned} A \cup B &= B_1 \cup B_2 \cup B_3 \\ P(A \cup B) &= P(B_1) + P(B_2) + P(B_3) \\ P(A) &= P(B_1) + P(B_2) \\ P(B) &= P(B_2) + P(B_3) \end{aligned}$$

Thus $P(A) + P(B) = P(B_1) + 2P(B_2) + P(B_3) = P(A \cup B) + P(A \cap B)$. □

1.5 Conditional Probability

Definition 1.5 (Conditional Probability).

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{provided } P(B) > 0$$

Example 1.6 (Dice Example). Roll two dice. Let A = "sum is 8", B = "first die is 3".

$$P(A|B) = \frac{P(\text{sum is 8 and first is 3})}{P(\text{first is 3})} = \frac{1/36}{1/6} = \frac{1}{6}$$

Theorem 1.7 (Multiplication Rule).

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

Theorem 1.8 (Law of Total Probability). If $\{B_i\}$ is a partition of Ω (disjoint and $\bigcup_i B_i = \Omega$), then:

$$P(A) = \sum_i P(A|B_i)P(B_i)$$

Theorem 1.9 (Bayes' Theorem).

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_j P(A|B_j)P(B_j)}$$

Example 1.10 (Medical Test). Disease prevalence = 1%, test sensitivity = 99%, specificity = 95%.

$$\begin{aligned} P(D) &= 0.01 \\ P(T|D) &= 0.99 \\ P(T|D^c) &= 0.05 \\ P(D|T) &= \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} \approx 0.167 \end{aligned}$$

Only 16.7% chance of actually having disease given positive test!

1.6 Independence

Definition 1.11 (Independence). Events A and B are independent if $P(A \cap B) = P(A)P(B)$.

Theorem 1.12 (Properties of Independence). 1. If A and B independent, then A and B^c , A^c and B , A^c and B^c are independent

2. For independent events $A_1, \dots, A_n$:

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i)$$

Definition 1.13 (Conditional Independence). A and B are conditionally independent given C if:

$$P(A \cap B|C) = P(A|C)P(B|C)$$

2 Random Variables

2.1 Definition and Classification

Definition 2.1 (Random Variable). A **random variable** X is a measurable function $X : \Omega \rightarrow \mathbb{R}$.

Definition 2.2 (Types of Random Variables). • **Discrete**: Takes countable values (finite or countably infinite)

- **Continuous**: Takes values in intervals (uncountably infinite)
- **Mixed**: Combination of discrete and continuous

2.2 Probability Mass Function (PMF)

Definition 2.3 (PMF). For discrete X , the PMF is:

$$p_X(x) = P(X = x)$$

with properties:

1. $p_X(x) \geq 0$ for all x
2. $\sum_x p_X(x) = 1$
3. $P(X \in A) = \sum_{x \in A} p_X(x)$

2.3 Probability Density Function (PDF)

Definition 2.4 (PDF). For continuous X , the PDF $f_X(x)$ satisfies:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

with properties:

1. $f_X(x) \geq 0$ for all x
2. $\int_{-\infty}^{\infty} f_X(x) dx = 1$
3. $P(X = x) = 0$ for all x

2.4 Cumulative Distribution Function (CDF)

Definition 2.5 (CDF).

$$F_X(x) = P(X \leq x) = \begin{cases} \sum_{t \leq x} p_X(t) & (\text{discrete}) \\ \int_{-\infty}^x f_X(t) dt & (\text{continuous}) \end{cases}$$

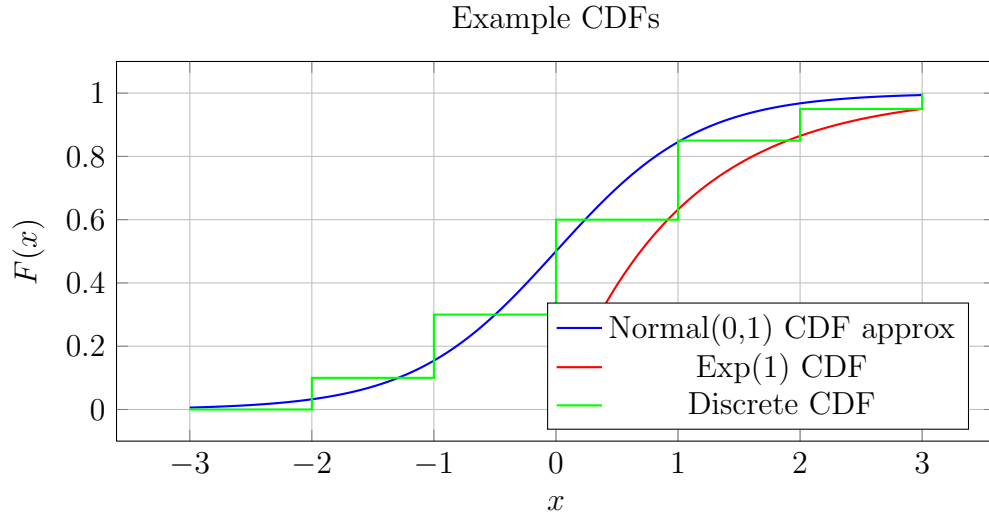


Figure 1: Examples of cumulative distribution functions

Theorem 2.6 (Properties of CDF). 1. *Non-decreasing:* $x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$

2. *Right-continuous:* $\lim_{y \rightarrow x^+} F(y) = F(x)$

3. $\lim_{x \rightarrow -\infty} F(x) = 0, \lim_{x \rightarrow \infty} F(x) = 1$

4. $P(a < X \leq b) = F(b) - F(a)$

5. $P(X > x) = 1 - F(x)$

2.5 Quantile Function

Definition 2.7 (Quantile). The p -th quantile x_p satisfies:

$$F(x_p) = p \quad \text{or} \quad P(X \leq x_p) \geq p \quad \text{and} \quad P(X \geq x_p) \geq 1 - p$$

2.6 Transformations of Random Variables

Theorem 2.8 (Transformations for Discrete RV). For $Y = g(X)$:

$$p_Y(y) = \sum_{x: g(x)=y} p_X(x)$$

Theorem 2.9 (Transformations for Continuous RV). *For $Y = g(X)$ with g monotonic and differentiable:*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

More generally, for $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$f_Y(\mathbf{y}) = f_X(g^{-1}(\mathbf{y})) |J_{g^{-1}}(\mathbf{y})|$$

where J is the Jacobian determinant.

Example 2.10 (Linear Transformation). *If $Y = aX + b$ with $a > 0$:*

$$f_Y(y) = \frac{1}{|a|} f_X\left(\frac{y-b}{a}\right)$$

3 Expectation and Moments

3.1 Expectation

Definition 3.1 (Expectation/Mean).

$$\mathbb{E}[X] = \begin{cases} \sum_x x p_X(x) & (\text{discrete}) \\ \int_{-\infty}^{\infty} x f_X(x) dx & (\text{continuous}) \end{cases}$$

provided the sum/integral converges absolutely.

Theorem 3.2 (Properties of Expectation). 1. *Linearity: $\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$*

2. *If $X \geq 0$, then $\mathbb{E}[X] \geq 0$*

3. *If $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$*

4. *Monotonicity: $X \geq 0$ and $\mathbb{E}[X] = 0$ implies $X = 0$ a.s.*

Theorem 3.3 (Law of the Unconscious Statistician (LOTUS)).

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x) p_X(x) & (\text{discrete}) \\ \int_{-\infty}^{\infty} g(x) f_X(x) dx & (\text{continuous}) \end{cases}$$

3.2 Variance and Standard Deviation

Definition 3.4 (Variance).

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Definition 3.5 (Standard Deviation).

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Theorem 3.6 (Properties of Variance). 1. $\text{Var}(aX + b) = a^2\text{Var}(X)$

2. $\text{Var}(X) \geq 0$

3. $\text{Var}(X) = 0$ iff X is constant almost surely

4. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

3.3 Moments

Definition 3.7 (Moments). The k -th moment: $\mu'_k = \mathbb{E}[X^k]$

The k -th central moment: $\mu_k = \mathbb{E}[(X - \mu)^k]$

where $\mu = \mathbb{E}[X]$.

Moment	Meaning
$\mu'_1 = \mu$	Mean (location)
$\mu_2 = \sigma^2$	Variance (spread)
μ_3	Related to skewness (asymmetry)
μ_4	Related to kurtosis (tailedness)
$\mu_1 = 0$	Always 0 (by definition)

Table 1: Interpretation of moments

3.4 Skewness

Definition 3.8 (Skewness).

$$\gamma_1 = \frac{\mu_3}{\sigma^3} = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}$$

- $\gamma_1 > 0$: Right-skewed (tail to right, mean > median > mode)
- $\gamma_1 < 0$: Left-skewed (tail to left, mean < median < mode)
- $\gamma_1 = 0$: Symmetric (approximately)

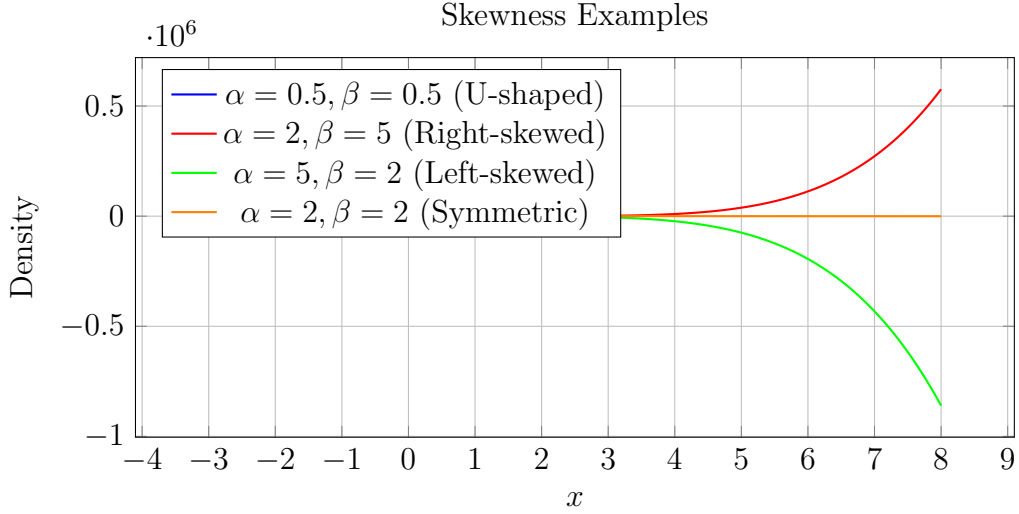


Figure 2: Distributions with different skewness

3.5 Kurtosis

Definition 3.9 (Kurtosis).

$$\gamma_2 = \frac{\mu_4}{\sigma^4} = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}$$

Excess kurtosis: $\gamma_2 - 3$

- $\gamma_2 > 3$: *Leptokurtic* (heavy tails, more outliers)
- $\gamma_2 = 3$: *Mesokurtic* (normal distribution)
- $\gamma_2 < 3$: *Platykurtic* (light tails, fewer outliers)

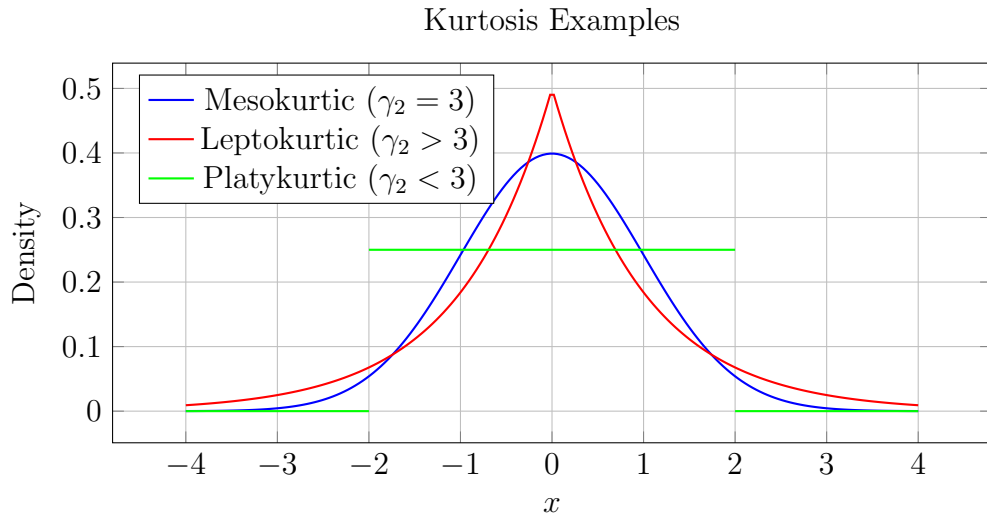


Figure 3: Distributions with different kurtosis

3.6 Covariance and Correlation

Definition 3.10 (Covariance).

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Definition 3.11 (Correlation).

$$\text{Corr}(X, Y) = \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

with $-1 \leq \rho_{XY} \leq 1$.

Theorem 3.12 (Properties of Covariance and Correlation). 1. *Symmetric:* $\text{Cov}(X, Y) = \text{Cov}(Y, X)$, $\text{Corr}(X, Y) = \text{Corr}(Y, X)$

2. *Bilinear:* $\text{Cov}(aX + bY, Z) = a\text{Cov}(X, Z) + b\text{Cov}(Y, Z)$

3. $\text{Cov}(X, X) = \text{Var}(X)$

4. $\text{Corr}(X, X) = 1$

5. $\text{Corr}(aX + b, cY + d) = \text{sign}(ac)\text{Corr}(X, Y)$

6. If X, Y independent, then $\text{Cov}(X, Y) = \text{Corr}(X, Y) = 0$ (converse not true)

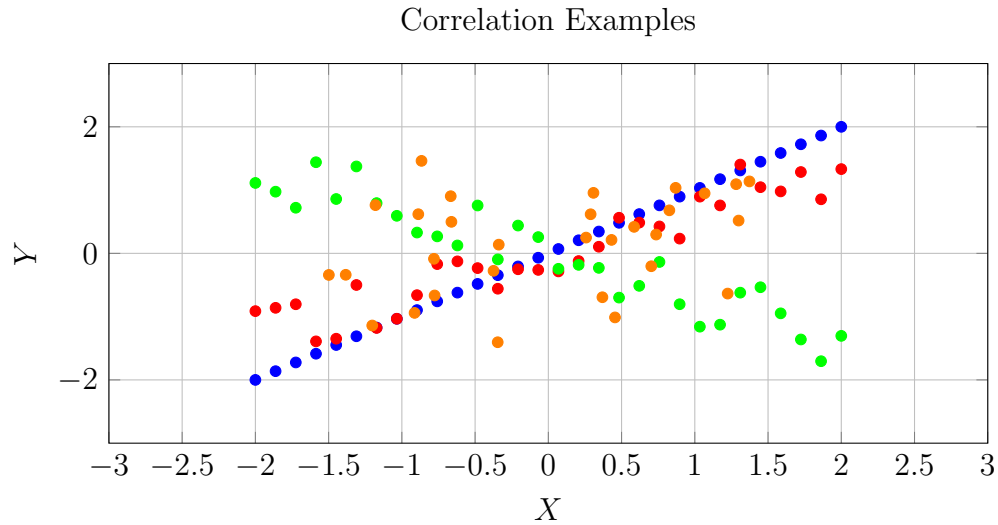


Figure 4: Different levels of correlation between variables

4 Common Discrete Distributions

4.1 Bernoulli Distribution

Definition 4.1 (Bernoulli Distribution). $X \sim \text{Ber}(p)$ models a single trial with success probability p :

$$p_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

- $\mathbb{E}[X] = p$
- $\text{Var}(X) = p(1 - p)$
- $M_X(t) = 1 - p + pe^t$

4.2 Binomial Distribution

Definition 4.2 (Binomial Distribution). $X \sim \text{Bin}(n, p)$ models number of successes in n independent Bernoulli trials:

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

- $\mathbb{E}[X] = np$
- $\text{Var}(X) = np(1 - p)$
- $M_X(t) = (1 - p + pe^t)^n$
- *Additive:* If $X \sim \text{Bin}(n, p)$, $Y \sim \text{Bin}(m, p)$ independent, then $X + Y \sim \text{Bin}(n + m, p)$

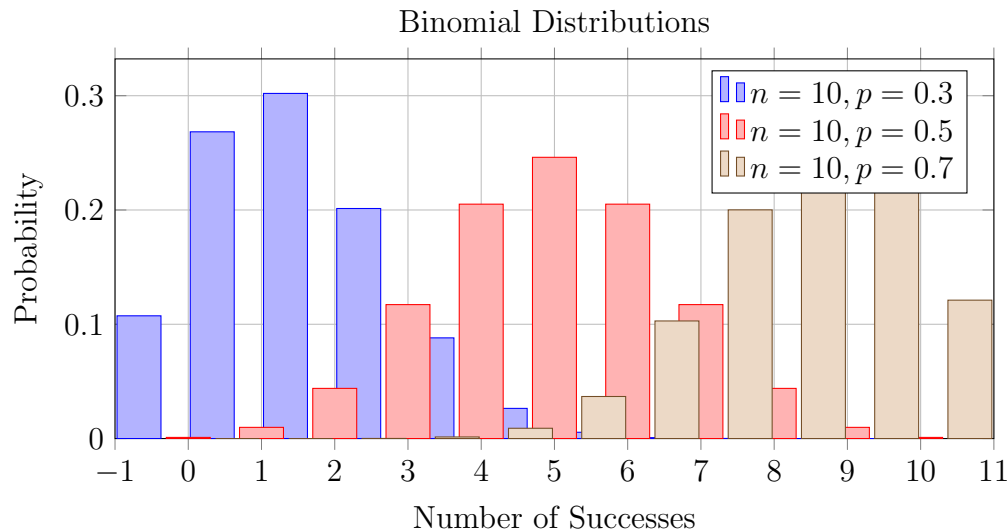


Figure 5: Binomial distributions with different success probabilities

4.3 Poisson Distribution

Definition 4.3 (Poisson Distribution). $X \sim \text{Pois}(\lambda)$ models number of events in fixed interval:

$$p_X(k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

- $\mathbb{E}[X] = \lambda$
- $\text{Var}(X) = \lambda$
- $M_X(t) = e^{\lambda(e^t - 1)}$
- *Additive:* If $X_i \sim \text{Pois}(\lambda_i)$ independent, then $\sum X_i \sim \text{Pois}(\sum \lambda_i)$

Theorem 4.4 (Poisson Approximation to Binomial). If $n \rightarrow \infty$, $p \rightarrow 0$, with $np \rightarrow \lambda$, then:

$$\text{Bin}(n, p) \xrightarrow{d} \text{Pois}(\lambda)$$

4.4 Geometric Distribution

Definition 4.5 (Geometric Distribution). $X \sim \text{Geo}(p)$ models number of trials until first success:

$$p_X(k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

- $\mathbb{E}[X] = \frac{1}{p}$
- $\text{Var}(X) = \frac{1-p}{p^2}$
- *Memoryless:* $P(X > n + m | X > m) = P(X > n)$

4.5 Negative Binomial Distribution

Definition 4.6 (Negative Binomial). $X \sim \text{NB}(r, p)$ models number of trials until r th success:

$$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \dots$$

- $\mathbb{E}[X] = \frac{r}{p}$
- $\text{Var}(X) = \frac{r(1-p)}{p^2}$
- *Sum of geometric:* If $X_i \sim \text{Geo}(p)$ i.i.d., then $\sum_{i=1}^r X_i \sim \text{NB}(r, p)$

5 Common Continuous Distributions

5.1 Uniform Distribution

Definition 5.1 (Uniform Distribution). $X \sim \text{Unif}(a, b)$ has constant density on $[a, b]$:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

- $\mathbb{E}[X] = \frac{a+b}{2}$
- $\text{Var}(X) = \frac{(b-a)^2}{12}$
- $F_X(x) = \frac{x-a}{b-a}$ for $x \in [a, b]$

5.2 Normal (Gaussian) Distribution

Definition 5.2 (Normal Distribution). $X \sim \mathcal{N}(\mu, \sigma^2)$ has PDF:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- $\mathbb{E}[X] = \mu$
- $\text{Var}(X) = \sigma^2$
- $M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$
- Symmetric about μ

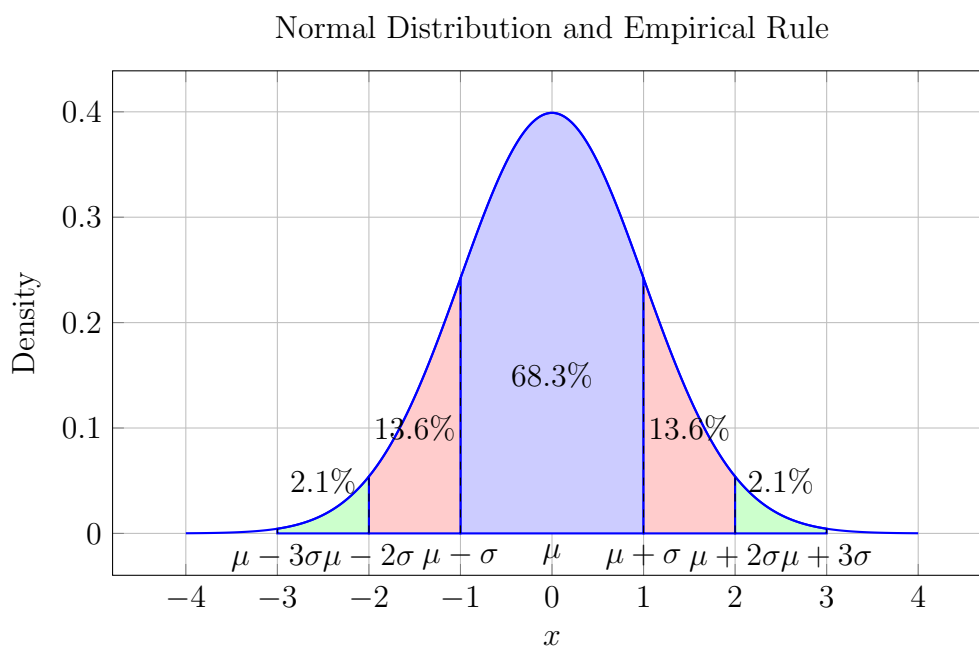


Figure 6: Normal distribution showing empirical rule (68-95-99.7 rule)

- Theorem 5.3** (Properties of Normal Distribution). 1. *Linear transformation: If $X \sim \mathcal{N}(\mu, \sigma^2)$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$*
2. *Standardization: $Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0, 1)$*
3. *Sum of normals: If $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ independent, then $X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$*
4. *If X, Y jointly normal, uncorrelated implies independent*

5.3 Exponential Distribution

Definition 5.4 (Exponential Distribution). $X \sim \text{Exp}(\lambda)$ models time between events in Poisson process:

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- $\mathbb{E}[X] = \frac{1}{\lambda}$
- $\text{Var}(X) = \frac{1}{\lambda^2}$
- $F_X(x) = 1 - e^{-\lambda x}$
- *Memoryless: $P(X > s + t | X > s) = P(X > t)$*

Theorem 5.5 (Relation to Poisson). *If events occur according to Poisson process with rate λ , then time between events is $\text{Exp}(\lambda)$.*

5.4 Gamma Distribution

Definition 5.6 (Gamma Distribution). $X \sim \text{Gamma}(\alpha, \beta)$ has PDF:

$$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$.

- $\mathbb{E}[X] = \frac{\alpha}{\beta}$
- $\text{Var}(X) = \frac{\alpha}{\beta^2}$
- $M_X(t) = \left(\frac{\beta}{\beta-t}\right)^\alpha$ for $t < \beta$
- $\text{Gamma}(1, \beta) = \text{Exp}(\beta)$
- $\text{Gamma}(n/2, 1/2) = \chi_n^2$ (Chi-square with n df)

5.5 Beta Distribution

Definition 5.7 (Beta Distribution). $X \sim \text{Beta}(\alpha, \beta)$ has PDF:

$$f_X(x) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1$$

where Γ is the gamma function.

- $\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}$
- $\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$
- Useful as prior for binomial proportion

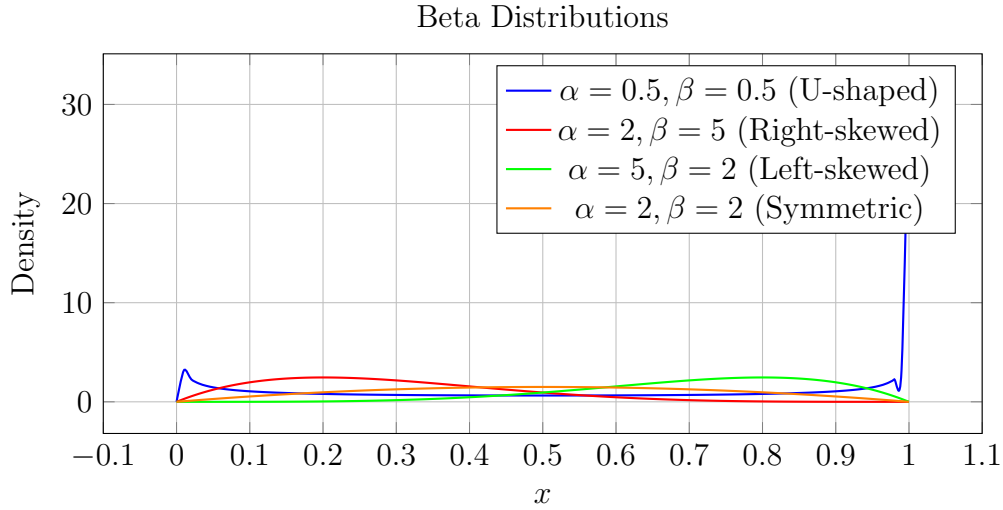


Figure 7: Beta distributions with different parameters

6 Joint Distributions and Dependence

6.1 Joint CDF and PDF

Definition 6.1 (Joint CDF). For random variables X, Y :

$$F_{XY}(x, y) = P(X \leq x, Y \leq y)$$

Definition 6.2 (Joint PDF). For continuous X, Y :

$$f_{XY}(x, y) = \frac{\partial^2 F_{XY}(x, y)}{\partial x \partial y}$$

with $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$.

6.2 Marginal Distributions

Definition 6.3 (Marginal PDF).

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

6.3 Conditional Distributions

Definition 6.4 (Conditional PDF).

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}, \quad f_Y(y) > 0$$

Theorem 6.5 (Bayes' Theorem for PDFs).

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{\int f_{Y|X}(y|x)f_X(x)dx}$$

6.4 Independence of Random Variables

Definition 6.6 (Independence). *X and Y are independent if:*

$$F_{XY}(x, y) = F_X(x)F_Y(y) \quad \text{or} \quad f_{XY}(x, y) = f_X(x)f_Y(y)$$

Theorem 6.7 (Properties of Independent RVs). *If X, Y independent:*

1. $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
2. $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
3. $M_{X+Y}(t) = M_X(t)M_Y(t)$
4. $\text{Cov}(X, Y) = 0$ (but converse not true)

6.5 Covariance Matrix

For random vector $\mathbf{X} = (X_1, \dots, X_n)^\top$:

Definition 6.8 (Covariance Matrix).

$$\Sigma = \text{Cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^\top]$$

where $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$ and $\Sigma_{ij} = \text{Cov}(X_i, X_j)$.

7 Limit Theorems

7.1 Convergence Concepts

Definition 7.1 (Convergence in Probability). $X_n \xrightarrow{P} X$ if $\forall \epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(|X_n - X| > \epsilon) = 0$$

Definition 7.2 (Almost Sure Convergence). $X_n \xrightarrow{a.s.} X$ if:

$$P\left(\lim_{n \rightarrow \infty} X_n = X\right) = 1$$

Definition 7.3 (Convergence in Distribution). $X_n \xrightarrow{d} X$ if:

$$\lim_{n \rightarrow \infty} F_{X_n}(x) = F_X(x)$$

for all x where F_X is continuous.

Theorem 7.4 (Relationships).

$$a.s. \text{ convergence} \Rightarrow \text{convergence in probability} \Rightarrow \text{convergence in distribution}$$

7.2 Law of Large Numbers

Theorem 7.5 (Weak Law of Large Numbers (WLLN)). Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{P} \mu$$

Theorem 7.6 (Strong Law of Large Numbers (SLLN)). Under same conditions as WLLN:

$$\bar{X}_n \xrightarrow{a.s.} \mu$$

7.3 Central Limit Theorem

Theorem 7.7 (Central Limit Theorem (CLT)). Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} Z \sim \mathcal{N}(0, 1)$$

Equivalently, for large n :

$$\bar{X}_n \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Idea of Proof using Characteristic Functions. Let $\phi(t) = \mathbb{E}[e^{itX}]$ be characteristic function. Then:

$$\begin{aligned}\phi_{\bar{X}_n}(t) &= \left[\phi\left(\frac{t}{n}\right) \right]^n \\ &= \left[1 + i\mu\frac{t}{n} - \frac{\sigma^2 t^2}{2n^2} + o\left(\frac{1}{n^2}\right) \right]^n \\ &\rightarrow e^{i\mu t - \frac{1}{2}\sigma^2 t^2} \quad \text{as } n \rightarrow \infty\end{aligned}$$

which is characteristic function of $\mathcal{N}(\mu, \sigma^2)$. □

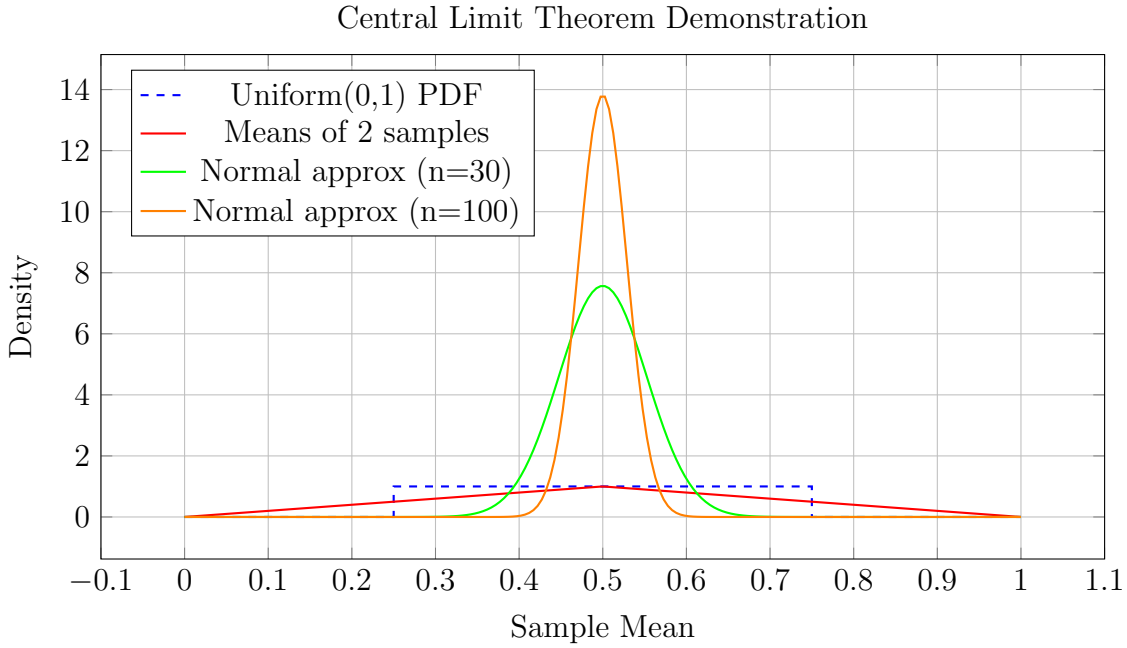


Figure 8: Central Limit Theorem in action: Distribution of sample means approaches normal as n increases

Theorem 7.8 (De Moivre-Laplace Theorem). *For $X \sim \text{Bin}(n, p)$, as $n \rightarrow \infty$ with $np(1-p) \rightarrow \infty$:*

$$\frac{X - np}{\sqrt{np(1-p)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

Continuity correction: For integers a, b :

$$P(a \leq X \leq b) \approx \Phi\left(\frac{b + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{a - 0.5 - np}{\sqrt{np(1-p)}}\right)$$

7.4 Delta Method

Theorem 7.9 (Delta Method). *If $\sqrt{n}(X_n - \theta) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then:*

$$\sqrt{n}(g(X_n) - g(\theta)) \xrightarrow{d} \mathcal{N}(0, [g'(\theta)]^2 \sigma^2)$$

8 Statistical Inference

8.1 Point Estimation

Definition 8.1 (Estimator). A statistic $\hat{\theta}_n = T(X_1, \dots, X_n)$ used to estimate parameter θ .

Definition 8.2 (Bias).

$$\text{Bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta$$

Definition 8.3 (Mean Squared Error).

$$\text{MSE}(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta}_n - \theta)^2] = \text{Var}(\hat{\theta}_n) + [\text{Bias}(\hat{\theta}_n)]^2$$

8.2 Properties of Estimators

Definition 8.4 (Consistency). $\hat{\theta}_n$ is consistent if $\hat{\theta}_n \xrightarrow{P} \theta$.

Definition 8.5 (Asymptotic Normality). $\hat{\theta}_n$ is asymptotically normal if:

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, V(\theta))$$

Definition 8.6 (Efficiency). An estimator is efficient if it achieves the Cramér-Rao lower bound.

8.3 Method of Moments

Method of Moments

Equate sample moments to population moments:

$$\frac{1}{n} \sum_{i=1}^n X_i^k = \mathbb{E}[X^k], \quad k = 1, 2, \dots, m$$

Solve m equations for m parameters.

8.4 Maximum Likelihood Estimation

Definition 8.7 (Likelihood Function).

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta)$$

Log-likelihood: $\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta)$.

MLE

Find $\hat{\theta}_{\text{MLE}}$ that maximizes $L(\theta)$ or $\ell(\theta)$:

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta \in \Theta} L(\theta) = \arg \max_{\theta \in \Theta} \ell(\theta)$$

Often solve score equation: $\frac{\partial \ell(\theta)}{\partial \theta} = 0$.

Example 8.8 (MLE for Normal Distribution). For $X_i \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d.:

$$\begin{aligned}\ell(\mu, \sigma^2) &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \\ \frac{\partial \ell}{\partial \mu} &= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = 0 \Rightarrow \hat{\mu} = \bar{x} \\ \frac{\partial \ell}{\partial \sigma^2} &= -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2\end{aligned}$$

Theorem 8.9 (Properties of MLE). Under regularity conditions:

1. Consistent: $\hat{\theta}_{\text{MLE}} \xrightarrow{P} \theta$
2. Asymptotically normal: $\sqrt{n}(\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1})$
3. Asymptotically efficient: Achieves Cramér-Rao lower bound
4. Invariant: If $\hat{\theta}$ is MLE of θ , then $g(\hat{\theta})$ is MLE of $g(\theta)$

8.5 Fisher Information

Definition 8.10 (Fisher Information). For single parameter θ :

$$I(\theta) = \mathbb{E} \left[\left(\frac{\partial}{\partial \theta} \log f(X; \theta) \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f(X; \theta) \right]$$

For sample of size n : $I_n(\theta) = nI(\theta)$.

Theorem 8.11 (Cramér-Rao Lower Bound). For any unbiased estimator $\hat{\theta}$:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{nI(\theta)}$$

8.6 Confidence Intervals

Definition 8.12 (Confidence Interval). *An interval $(L(X), U(X))$ such that:*

$$P(\theta \in (L(X), U(X))) = 1 - \alpha$$

Theorem 8.13 (Common Confidence Intervals). 1. **Normal mean, known variance**
(σ^2 known):

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

2. **Normal mean, unknown variance:**

$$\bar{X} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

3. **Normal variance:**

$$\left(\frac{(n-1)s^2}{\chi_{n-1, \alpha/2}^2}, \frac{(n-1)s^2}{\chi_{n-1, 1-\alpha/2}^2} \right)$$

4. **Binomial proportion** (approximate):

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

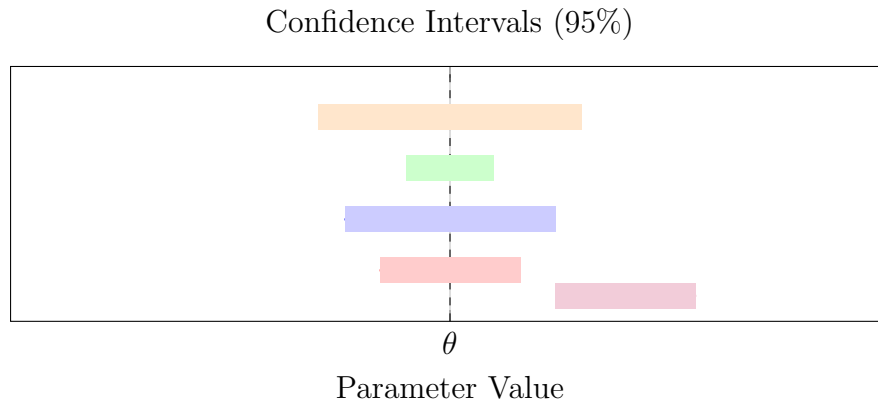


Figure 9: Interpretation of confidence intervals: 95% of intervals should contain the true parameter

8.7 Hypothesis Testing

Definition 8.14 (Hypothesis Test Components). • *Null hypothesis H_0 (status quo)*

- *Alternative hypothesis H_1 or H_a (research hypothesis)*
- *Test statistic $T(X)$*

- *Rejection region R*
- *Significance level $\alpha = P(\text{Type I error})$*
- *Power $1 - \beta = P(\text{Reject } H_0 | H_1 \text{ true})$*

Definition 8.15 (Errors in Hypothesis Testing).

Decision	True State of Nature	
	H_0 true	H_1 true
Reject H_0	Type I error (α)	Correct decision (Power)
Fail to reject H_0	Correct decision	Type II error (β)

Theorem 8.16 (Neyman-Pearson Lemma). *For simple hypotheses $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$, the most powerful test rejects when:*

$$\frac{L(\theta_1)}{L(\theta_0)} > k$$

where k is chosen such that $P(\text{Reject } H_0 | H_0) = \alpha$.

8.8 p-value

Definition 8.17 (p-value). *The probability, under H_0 , of obtaining a test statistic at least as extreme as the observed value.*

$$p\text{-value} = \inf\{\alpha : \text{reject } H_0 \text{ at level } \alpha\}$$

Remark 8.18. • *Small p -value $\Rightarrow$ evidence against H_0*

- *$p\text{-value} < \alpha \Rightarrow$ reject H_0 at level α*
- *p -value is NOT probability H_0 is true*

8.9 Common Tests

8.9.1 z-test

For $H_0 : \mu = \mu_0$, known σ :

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1) \text{ under } H_0$$

8.9.2 t-test

For $H_0 : \mu = \mu_0$, unknown σ :

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0$$

8.9.3 Chi-square Test for Variance

For $H_0 : \sigma^2 = \sigma_0^2$ (normal population):

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} \sim \chi_{n-1}^2 \text{ under } H_0$$

8.9.4 F-test for Equality of Variances

For $H_0 : \sigma_1^2 = \sigma_2^2$ (two normal populations):

$$F = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1} \text{ under } H_0$$

8.9.5 Chi-square Test for Independence

For contingency table with r rows, c columns:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi_{(r-1)(c-1)}^2$$

where $E_{ij} = \frac{\text{row total}_i \times \text{col total}_j}{n}$.

8.10 ANOVA (Analysis of Variance)

Theorem 8.19 (One-way ANOVA). *Test $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$:*

$$F = \frac{MSB}{MSW} \sim F_{k-1, n-k} \text{ under } H_0$$

where:

$$SSB = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X})^2 \quad (\text{Between groups})$$

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2 \quad (\text{Within groups})$$

$$SST = SSB + SSW \quad (\text{Total})$$

$$MSB = \frac{SSB}{k-1}, \quad MSW = \frac{SSW}{n-k}$$

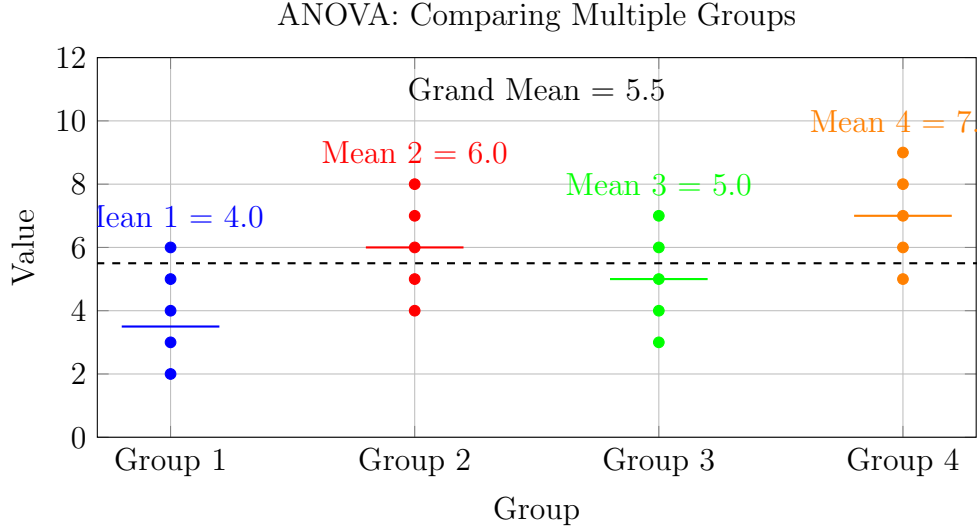


Figure 10: ANOVA decomposes total variation into between-group and within-group variation

9 Regression Analysis

9.1 Simple Linear Regression

Definition 9.1 (Simple Linear Model).

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, \dots, n$$

where $\epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$.

Theorem 9.2 (Ordinary Least Squares (OLS) Estimators). *Minimize sum of squared errors:*

$$SSE = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

Solutions:

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = \frac{s_{XY}}{s_X^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Theorem 9.3 (Properties of OLS Estimators). *Under assumptions:*

1. *Unbiased:* $\mathbb{E}[\hat{\beta}_0] = \beta_0$, $\mathbb{E}[\hat{\beta}_1] = \beta_1$

2. *Variances:*

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum (X_i - \bar{X})^2}$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum (X_i - \bar{X})^2} \right)$$

3. Covariance: $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{X}}{\sum (X_i - \bar{X})^2}$

4. $\hat{\beta}_0, \hat{\beta}_1$ are BLUE (Best Linear Unbiased Estimators, Gauss-Markov)

9.2 Multiple Linear Regression

Definition 9.4 (Multiple Linear Model).

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where:

- $\mathbf{Y} \in \mathbb{R}^n$: response vector
- $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$: design matrix (includes intercept)
- $\boldsymbol{\beta} \in \mathbb{R}^{p+1}$: parameter vector
- $\boldsymbol{\epsilon} \in \mathbb{R}^n$: error vector, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$

Theorem 9.5 (OLS Solution).

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

provided $\mathbf{X}^\top \mathbf{X}$ is invertible.

Theorem 9.6 (Properties of Multiple Regression OLS). 1. $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$

2. $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$

3. $\hat{\boldsymbol{\beta}} \sim \mathcal{N}(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$

9.3 Coefficient of Determination

Definition 9.7 (R^2 and Adjusted R^2).

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

$$R_{adj}^2 = 1 - \frac{SSE/(n-p-1)}{SST/(n-1)} = 1 - (1 - R^2) \frac{n-1}{n-p-1}$$

where:

$$SST = \sum (Y_i - \bar{Y})^2 \quad (\text{Total sum of squares})$$

$$SSR = \sum (\hat{Y}_i - \bar{Y})^2 \quad (\text{Regression sum of squares})$$

$$SSE = \sum (Y_i - \hat{Y}_i)^2 \quad (\text{Error sum of squares})$$

9.4 Diagnostics and Assumptions

Key assumptions for linear regression:

1. Linearity: $E[Y|X] = \beta_0 + \beta_1 X$
2. Independence: Errors ϵ_i are independent
3. Homoscedasticity: Constant variance $\text{Var}(\epsilon_i) = \sigma^2$
4. Normality: $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$
5. No perfect multicollinearity: $\mathbf{X}$ has full column rank

10 Bayesian Statistics

10.1 Bayes' Theorem for Parameters

$$\pi(\theta|\mathbf{x}) = \frac{f(\mathbf{x}|\theta)\pi(\theta)}{\int f(\mathbf{x}|\theta)\pi(\theta)d\theta} \propto f(\mathbf{x}|\theta)\pi(\theta)$$

- $\pi(\theta)$: Prior distribution (belief before data)
- $f(\mathbf{x}|\theta)$: Likelihood function
- $\pi(\theta|\mathbf{x})$: Posterior distribution (belief after data)
- $f(\mathbf{x})$: Marginal likelihood/evidence

10.2 Conjugate Priors

Likelihood	Parameter	Conjugate Prior
Binomial	p	Beta(α, β)
Poisson	λ	Gamma(α, β)
Normal (known σ^2)	μ	Normal(μ_0, τ_0^2)
Normal (known μ)	σ^2	Inverse-Gamma(α, β)
Multinomial	$\mathbf{p}$	Dirichlet($\boldsymbol{\alpha}$)

Table 2: Common conjugate prior families

10.3 Bayesian Inference

Definition 10.1 (Posterior Mean).

$$\mathbb{E}[\theta|\mathbf{x}] = \int \theta \pi(\theta|\mathbf{x}) d\theta$$

Definition 10.2 (Credible Interval). (a, b) such that:

$$P(a \leq \theta \leq b|\mathbf{x}) = 1 - \alpha$$

10.4 Bayesian Hypothesis Testing

Definition 10.3 (Bayes Factor). *For comparing $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$:*

$$BF_{01} = \frac{P(\mathbf{x}|H_0)}{P(\mathbf{x}|H_1)} = \frac{\int_{\Theta_0} f(\mathbf{x}|\theta)\pi_0(\theta)d\theta}{\int_{\Theta_1} f(\mathbf{x}|\theta)\pi_1(\theta)d\theta}$$

Bayes Factor	Evidence against H_0
$BF > 100$	Decisive
$30 < BF \leq 100$	Very strong
$10 < BF \leq 30$	Strong
$3 < BF \leq 10$	Substantial
$1 < BF \leq 3$	Barely worth mentioning

Table 3: Interpretation of Bayes factors

11 Nonparametric Statistics

11.1 Empirical Distribution Function

Definition 11.1 (EDF).

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$$

Theorem 11.2 (Glivenko-Cantelli Theorem).

$$\sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0$$

Theorem 11.3 (Donsker's Theorem).

$$\sqrt{n}(\hat{F}_n(x) - F(x)) \xrightarrow{d} \mathbb{B}(F(x))$$

where $\mathbb{B}$ is Brownian bridge.

11.2 Kernel Density Estimation

Definition 11.4 (Kernel Density Estimator).

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

Common kernels:

- *Gaussian:* $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$
- *Epanechnikov:* $K(u) = \frac{3}{4}(1 - u^2)I(|u| \leq 1)$
- *Uniform:* $K(u) = \frac{1}{2}I(|u| \leq 1)$

11.3 Bootstrap Methods

Nonparametric Bootstrap

1. Draw B bootstrap samples $\mathbf{X}^{*1}, \dots, \mathbf{X}^{*B}$ from $\hat{F}_n$
2. Compute $\hat{\theta}^{*b} = T(\mathbf{X}^{*b})$ for each
3. Use $\{\hat{\theta}^{*1}, \dots, \hat{\theta}^{*B}\}$ to estimate:
 - Bias: $\widehat{\text{Bias}} = \bar{\theta}^* - \hat{\theta}$
 - Variance: $\widehat{\text{Var}} = \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{*b} - \bar{\theta}^*)^2$
 - Confidence intervals

Theorem 11.5 (Bootstrap Consistency). *Under regularity conditions, bootstrap approximates sampling distribution:*

$$\sqrt{n}(\hat{\theta}_n^* - \hat{\theta}_n) \xrightarrow{d} \sqrt{n}(\hat{\theta}_n - \theta)$$

12 Time Series Analysis

12.1 Stationarity

Definition 12.1 (Stationarity). *Time series $\{X_t\}$ is:*

- *Strictly stationary if $(X_{t_1}, \dots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, \dots, X_{t_k+h})$*
- *Weakly stationary if:*
 1. $\mathbb{E}[X_t] = \mu$ constant
 2. $\text{Cov}(X_t, X_{t+h}) = \gamma(h)$ depends only on lag h

12.2 Autocorrelation Function

Definition 12.2 (ACF and PACF). *Autocorrelation function:*

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{\text{Cov}(X_t, X_{t+h})}{\text{Var}(X_t)}$$

Partial autocorrelation function ϕ_{hh} : Correlation between X_t and X_{t+h} after removing linear dependence on $X_{t+1}, \dots, X_{t+h-1}$.

12.3 ARIMA Models

Definition 12.3 (ARIMA(p,d,q)).

$$(1 - \phi_1 B - \dots - \phi_p B^p)(1 - B)^d X_t = (1 + \theta_1 B + \dots + \theta_q B^q) \epsilon_t$$

where B is backshift operator: $BX_t = X_{t-1}$, and $\epsilon_t \sim WN(0, \sigma^2)$.

Model	Equation	Properties
AR(1)	$X_t = \phi X_{t-1} + \epsilon_t$	$\rho(h) = \phi^{ h }$
MA(1)	$X_t = \epsilon_t + \theta \epsilon_{t-1}$	$\rho(1) = \frac{\theta}{1+\theta^2}, \rho(h) = 0$ for $ h > 1$
ARMA(1,1)	$X_t = \phi X_{t-1} + \epsilon_t + \theta \epsilon_{t-1}$	More flexible decay pattern

Table 4: Common time series models

13 Exercises

13.1 Basic Exercises

Exercise 13.1. *Prove that $\text{Var}(aX + b) = a^2 \text{Var}(X)$.*

Exercise 13.2. *Find MLE for λ in $\text{Pois}(\lambda)$ distribution.*

Exercise 13.3. *Derive 95% confidence interval for normal mean with unknown variance.*

13.2 Intermediate Exercises

Exercise 13.4. *Prove Central Limit Theorem using characteristic functions.*

Exercise 13.5. *Derive OLS estimators for simple linear regression.*

Exercise 13.6. *Find posterior distribution for normal mean with normal prior.*

13.3 Advanced Exercises

Exercise 13.7. *Prove Neyman-Pearson Lemma.*

Exercise 13.8. *Derive Cramér-Rao lower bound.*

Exercise 13.9. *Implement bootstrap method for confidence intervals.*

Appendix: Statistical Tables

Standard Normal Distribution

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
-3.00	0.0013	-1.00	0.1587	1.00	0.8413
-2.58	0.0049	-0.67	0.2514	1.28	0.8997
-2.33	0.0099	-0.50	0.3085	1.50	0.9332
-2.00	0.0228	-0.25	0.4013	1.64	0.9495
-1.96	0.0250	0.00	0.5000	1.96	0.9750
-1.64	0.0505	0.25	0.5987	2.00	0.9772
-1.50	0.0668	0.50	0.6915	2.33	0.9901
-1.28	0.1003	0.67	0.7486	2.58	0.9951

Table 5: Cumulative probabilities for standard normal distribution $\Phi(z) = P(Z \leq z)$

Critical Values for t-distribution

df	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.025$	$\alpha = 0.01$	$\alpha = 0.005$
1	3.078	6.314	12.706	31.821	63.657
5	1.476	2.015	2.571	3.365	4.032
10	1.372	1.812	2.228	2.764	3.169
20	1.325	1.725	2.086	2.528	2.845
30	1.310	1.697	2.042	2.457	2.750
60	1.296	1.671	2.000	2.390	2.660
120	1.289	1.658	1.980	2.358	2.617
∞	1.282	1.645	1.960	2.326	2.576

Table 6: Critical values for t-distribution (two-tailed)

Critical Values for Chi-square Distribution

df	$\alpha = 0.99$	$\alpha = 0.975$	$\alpha = 0.95$	$\alpha = 0.05$	$\alpha = 0.025$
1	0.000	0.001	0.004	3.841	5.024
5	0.554	0.831	1.145	11.070	12.833
10	2.558	3.247	3.940	18.307	20.483
20	8.260	9.591	10.851	31.410	34.170
30	14.953	16.791	18.493	43.773	46.979

Table 7: Critical values for χ^2 distribution

Common Probability Distributions Summary

Distribution	PDF/PMF	Mean	Variance	MGF/Characteristic Function
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$	$1-p+pe^t$
Binomial(n, p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$(1-p+pe^t)^n$
Poisson(λ)	$\frac{\lambda^k e^{-\lambda}}{k!}$	λ	λ	$e^{\lambda(e^t-1)}$
Geometric(p)	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\frac{pe^t}{1-(1-p)e^t}$
Uniform(a, b)	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$e^{\mu t + \frac{1}{2}\sigma^2 t^2}$
Exponential(λ)	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{\lambda}{\lambda-t}$
Gamma(α, β)	$\frac{\beta^\alpha}{\Gamma(\alpha)}x^{\alpha-1}e^{-\beta x}$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\left(\frac{\beta}{\beta-t}\right)^\alpha$
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$F_1(\alpha; \alpha+\beta; t)$

Table 8: Summary of common probability distributions

Statistical Formulas Cheat Sheet

Sample Mean: $\bar{x} = \frac{1}{n} \sum x_i$

Sample Variance: $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

Sample Covariance: $s_{xy} = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$

Sample Correlation: $r = \frac{s_{xy}}{s_x s_y}$

Sample Skewness: $g_1 = \frac{\frac{1}{n} \sum (x_i - \bar{x})^3}{s^3}$

Sample Kurtosis: $g_2 = \frac{\frac{1}{n} \sum (x_i - \bar{x})^4}{s^4}$

Regression Slope: $b_1 = \frac{s_{xy}}{s_x^2}$

Regression Intercept: $b_0 = \bar{y} - b_1 \bar{x}$

Coefficient of Determination: $R^2 = \frac{s_{xy}^2}{s_x^2 s_y^2}$

Standard Error of Mean: $SE_{\bar{x}} = \frac{s}{\sqrt{n}}$

Margin of Error: $ME = z_{\alpha/2} \frac{s}{\sqrt{n}}$

Test Statistic (z): $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

Test Statistic (t): $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

Chi-square Statistic: $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$

F-statistic: $F = \frac{s_1^2}{s_2^2}$

Confidence Interval: $\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$

Probability Rules Summary

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$$P(A^c) = 1 - P(A)$$

$$P(A \cap B^c) = P(A) - P(A \cap B)$$

$$\text{If } A \subseteq B \text{ then } P(A) \leq P(B)$$

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i) \quad (\text{Boole})$$

$$P\left(\bigcap_{i=1}^n A_i\right) \geq 1 - \sum_{i=1}^n P(A_i^c) \quad (\text{Bonferroni})$$

Expectation and Variance Rules

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

$$\text{Var}(aX + b) = a^2\text{Var}(X)$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$\mathbb{E}[X^2] = \text{Var}(X) + (\mathbb{E}[X])^2$$

$$\text{Cov}(aX + b, cY + d) = ac\text{Cov}(X, Y)$$

$$\text{If } X, Y \text{ independent: } \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$

$$\text{If } X, Y \text{ independent: } \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Common Limits and Approximations

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x$$

$$\text{Stirling: } n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

Poisson approx to Binomial: $\text{Bin}(n, p) \approx \text{Pois}(np)$ if n large, p small

Normal approx to Binomial: $\text{Bin}(n, p) \approx \mathcal{N}(np, np(1-p))$ if $np(1-p) > 5$

Central Limit Theorem: $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1)$ for large n

Delta Method: $\sqrt{n}(g(\bar{X}) - g(\mu)) \approx \mathcal{N}(0, [g'(\mu)]^2 \sigma^2)$